

QUALITY SCORES

How to Identify Financially Strong Companies With the Piotroski F-Score



July 11, 2024 ■ 27 min read

Share this article

Article font size

A A

Table of Contents

- Piotroski F-Score Explained
- Piotroski F-Score Components
- How to Interpret the Piotroski F-Score Ranges
- Piotroski F-Score Calculation Example
 - Step #1: Gather Necessary Financial Data
 - Step #2: Calculate Relevant Ratios
 - Step #3: Determine Piotroski's F-Score
- Piotroski's F-Score and Investment Performance
- Limitations of the Piotroski F-Score
- The Bottom Line

In this article, I will show you how to identify financially strong companies using the Piotroski F-Score. This nine-point scoring system, developed by accounting professor Joseph D. Piotroski in 2000, evaluates a firm's financial health and provides a quantitative framework for distinguishing between high-quality and low-quality value stocks. In essence, Piotroski's approach identifies high-quality stocks as those that are profitable, have improving margins, don't employ accounting tricks, and demonstrate strengthening balance sheets.

This article will explain the Piotroski F-Score, its calculation, and interpretation. We'll provide a real-world example, a free Excel template, and discuss Piotroski's research on the score's investment performance. We'll also cover studies that support and add to the F-Score, as well as its key limitations and a study suggesting it may not hold up well in practice.

Piotroski F-Score Explained

[Joseph D. Piotroski](#), an Associate Professor of Accounting at the Stanford University Graduate School of Business, developed the **Piotroski F-Score** in his influential paper "[Value Investing: The Use of Historical Financial Statement Information to Separate Winners from Losers](#)." The paper, published in 2000 while he was an Associate Professor of Accounting at the University of Chicago's Graduate School of Business, presented the F-Score as a method for assessing a company's financial strength using solely historical financial statement data.

Piotroski's research focused on identifying high book-to-market firms (meaning the market is valuing the company's equity cheaply compared to its net assets) that were financially strong and likely to outperform the market. He argued that through a simple accounting-based fundamental analysis strategy, investors could distinguish the more promising companies from the less promising ones among these firms.

As the opening sentence of Piotroski's paper states, the central thesis of his research was:

"This paper examines whether a simple accounting-based fundamental analysis strategy, when applied to a broad portfolio of high book-to-market firms, can shift the distribution of returns earned by an investor."

- Joseph D. Piotroski in "Value Investing: The Use Of Historical Financial Statement Information To Separate Winners From Losers," pg. 1

The results of his study were compelling. Piotroski demonstrated that by focusing on financially strong firms within the high book-to-market (BM) group, as identified by the F-Score, investors can significantly improve their investment performance and shift the distribution of their returns to the right.

The second and third sentences of the paper summarize the key findings:

"I show that the mean return earned by a high book-to-market investor can be increased by **at least 7½% annually** through the selection of financially strong high BM firms while the entire distribution of realized returns is shifted to the right. In addition, an investment strategy that buys expected winners and shorts expected losers generates a **23% annual return between 1976 and 1996**, and the strategy appears to be robust across time and to controls for alternative investment strategies."

- Joseph D. Piotroski in "Value Investing: The Use Of Historical Financial Statement Information To Separate Winners From Losers," pg. 1

These findings, as we'll explore closely later in this article, highlight the potential of the F-Score as a tool for investors to enhance their returns when investing in high book-to-market companies.

Piotroski F-Score Components

The F-Score is calculated based on nine financial criteria (what Piotroski calls "binary signals") related to a firm's profitability; leverage, liquidity, source of funds; and operating efficiency. Each criterion is assigned a value of either 1 (good) or 0 (bad), and the F-Score is the sum of these nine values, ranging from 0 to 9. A higher F-Score indicates a firm with stronger financial position.

The Piotroski F-Score is broken down into the following categories:

1. Profitability.
2. Leverage, liquidity, and source of funds.
3. Operating efficiency.

Below are the nine Piotroski F-Score factors separated into their respective categories, detailing their purpose and academic notations (in parentheses), along with a brief explanation for each:

Profitability

1. **Positive Return on Assets (F_{ROA})**: Measures whether the firm is generating positive return on assets (ROA) (net income / total assets). A positive ROA indicates the firm is profitable and efficiently utilizing its assets. 1 if ROA is positive, 0 otherwise.
2. **Positive Operating Cash Flow (F_{CFO})**: Assesses whether the firm is generating positive cash flow from operations. Positive CFO indicates the firm's core business is generating cash. 1 if CFO is positive, 0 otherwise.
3. **Improving Return on Assets (F_{ΔROA})**: Captures whether the firm's profitability is improving compared to the previous year. As noted in Piotroski's paper, a positive earnings trend is suggestive of an improvement in the firm's underlying ability to generate positive future cash flows. 1 if ROA improved compared to the previous year, 0 otherwise.
4. **Quality of Earnings (F_{ACCRUAL})**: Compares the firm's cash flow from operations (CFO) to its return on assets (ROA). If the ratio of CFO to total assets (aka cash return on assets) exceeds ROA in the current year, it suggests the firm's earnings are of high quality and are backed by cash flows. Piotroski's paper highlights that this relationship may be particularly important among value firms, where the incentive to manage earnings through positive accruals (e.g., to prevent covenant violations) is strong. 1 if CFO/Total Assets > Net Income/Total Assets (ROA), 0 otherwise.

The first four criteria in the Piotroski F-Score assess a company's profitability. Profitable firms with positive net income and operating cash flow are more likely to have the financial resources to weather economic downturns and capitalize on growth opportunities.

Leverage, Liquidity, and Source of Funds

5. **Decreasing Leverage (F_{ΔLEVER})**: Measures changes in the firm's long-term debt levels relative to total assets (long-term debt / total assets). A decrease in the leverage ratio is viewed positively as it reduces financial risk. Piotroski's paper notes that since most high BM firms are financially constrained, an increase in leverage is likely to place additional constraints on the firm's financial flexibility. 1 if the leverage ratio decreased compared to the previous year, 0 otherwise.
6. **Increasing Liquidity (F_{ΔLIQUID})**: Captures changes in the firm's current ratio (current assets / current liabilities). An improvement in liquidity is a good signal about the firm's ability to meet short-term obligations. 1 if the current ratio

Financially distressed firms often raise equity capital as a last resort. Piotroski's paper notes that by raising external capital, a financially distressed firm is signaling its inability to generate sufficient internal funds. 1 if the firm did not issue equity in the previous year, 0 otherwise.

The next three criteria evaluate a company's financial leverage, liquidity, and source of funding. Firms with decreasing debt levels, stable or improving current ratios, and a lack of new equity issuances are generally in a stronger financial position, as they are less reliant on external financing and have more flexibility to invest in their operations.

Operating Efficiency

- 8. **Improving Gross Margin (F_ΔMARGIN):** Measures the change in the firm's gross margin compared to the previous year. An improvement in margins suggests better operating efficiency, cost control, or pricing power. 1 if the gross margin improved compared to the previous year, 0 otherwise.
- 9. **Improving Asset Turnover (F_ΔTURN):** Captures the change in the firm's asset turnover ratio (revenues / total assets). An improvement in asset turnover indicates the firm is generating more sales per dollar of assets, suggesting better efficiency. Piotroski's paper highlights that an improvement in asset turnover can arise from more efficient operations (fewer assets generating the same levels of sales) or an increase in sales (which could also signify improved market conditions for the firm's products). 1 if the asset turnover ratio improved compared to the previous year, 0 otherwise.

The final two criteria measure a company's operating efficiency, as reflected in its gross margin and asset turnover ratio. Improvements in these metrics can indicate that a firm is becoming more productive and efficient in its use of resources, which can lead to higher profitability and shareholder returns.

How to Interpret the Piotroski F-Score Ranges

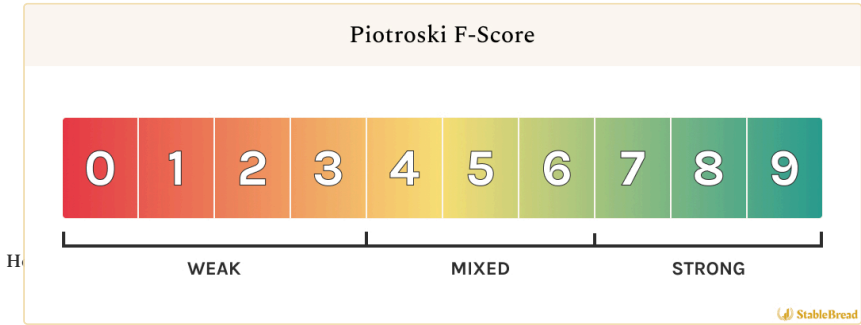
The Piotroski F-Score is an aggregate measure of a firm's financial strength, calculated using nine criteria based on profitability; leverage, liquidity, source of funds; and operating efficiency. Each criterion scores 1 if met (positive signal) or 0 if not met, and the individual scores are summed to calculate the final F-Score, ranging from 0 (weakest) to 9 (strongest).

Thus, the Piotroski F-Score can be calculated using the following formula:

F_SCORE = F_ROA + F_ΔROA + F_CFO + F_ACCRUAL + F_ΔMARGIN + F_ΔTURN + F_ΔLEVER + F_ΔLIQUID + EQ_OFFER

In short, Piotroski's paper suggests that investors can benefit from prioritizing high book-to-market firms with higher F-Scores and avoiding those with lower F-Scores.

For a high-level interpretation of the Piotroski F-Score ranges, the three-bucket visualization below can be referenced:



Piotroski F-Score | StableBread

Subscribe to the Email List!

Receive updates on articles, website tools, spreadsheets, and everything value-investing related.

Email *

☐ I consent to my information being stored and used to receive the StableBread newsletter *

- 1. Gather the necessary financial data for the company, including the current and prior years income statements, balance sheets, and cash flow statements.
- 2. Calculate the relevant ratios to determine whether the nine criteria score a 1 or 0.
- 3. Evaluate the company against each of the nine criteria and assign a point for every criterion that is met. Sum the total points to arrive at the company's Piotroski F-Score, which will range from 0 to 9.

To illustrate how to calculate the Piotroski F-Score, let's consider the example of Lowe's (LOW), an American retail company specializing in home improvement.

You can use the spreadsheet model linked below to complete the Piotroski F-Score calculation, which is effectively what steps #1-3 below cover. It just requires you to input the company's last reported fiscal year, then all the relevant financials (in the yellow input cells). Everything else will be calculated for you:

Piotroski F-Score - StableBread Download

Step #1: Gather Necessary Financial Data

To calculate the Piotroski F-Score for any company, you'll need to gather relevant financial data from the income statement, balance sheet, and cash flow statement. Financial statements can be sourced from the company's 10-K annual report, which can be found via the SEC or the company's investor relations pages.

specific period.

Featured Tool

Unlock smarter investing with StableBread's Automated Stock Analysis Spreadsheet. Effortlessly analyze company fundamentals, financial statements, and valuations. No manual data collection required. From the income statement, you'll need to locate these three financials:

Learn More

- **Related:** [How to Effectively Read and Analyze an Income Statement](#)
- **Revenues:** For the asset turnover and gross profit margin calculations.
- **Gross Profit:** To calculate gross profit margin.
- **Net Income:** To evaluate whether it's positive and for the return on assets calculation.

Here's Lowe's income statement with these three financials outlined:

Lowe's Companies, Inc. Consolidated Statements of Earnings (In millions, except per share and percentage data)							
	February 2, 2024		Fiscal Years Ended February 3, 2023		January 28, 2022		
	Amount	% Sales	Amount	% Sales	Amount	% Sales	
Current Earnings							
Net sales	\$ 86,377	100.00 %	\$ 97,059	100.00 %	\$ 86,250	100.00 %	
Cost of sales	57,533	66.61	64,802	66.77	64,194	66.70	
Gross margin	28,844	33.39	32,257	33.23	32,056	33.30	
Expenses:							
Selling, general and administrative	15,576	18.02	20,352	20.94	18,301	19.01	
Depreciation and amortization	1,717	1.99	1,766	1.82	1,662	1.73	
Operating income	11,557	13.38	10,159	10.47	12,093	12.56	
Interest - net	1,382	1.60	1,123	1.16	885	0.92	
Pre-tax earnings	10,175	11.78	9,036	9.31	11,208	11.64	
Income tax provision	2,449	2.83	2,599	2.68	2,766	2.87	
Net earnings	\$ 7,726	8.95 %	\$ 6,437	6.63 %	\$ 8,442	8.77 %	
Basic earnings per common share	\$ 13.23		\$ 10.20		\$ 12.07		
Diluted earnings per common share	\$ 13.20		\$ 10.17		\$ 12.04		

Source: [Lowe's \(LOW\) 10-K Annual Statement](#) (Income Statement)

Thus, in FY 2024, Lowe's reported revenues of \$86,377M, gross profit of \$28,844M, and net income of \$7,726M.

Balance Sheet

The balance sheet provides a snapshot of a company's financial position at a specific point in time, detailing assets, liabilities, and shareholders' equity.

- **Related:** [How to Effectively Read and Analyze a Balance Sheet](#)

From the balance sheet, you'll need to locate these four financials:

- **Current Assets:** To calculate the current ratio.
- **Total Assets:** For the return on assets, cash return on assets, long-term debt to total assets, and asset turnover ratio calculations.
- **Current Liabilities:** Also to calculate the current ratio.
- **Long-Term Debt:** To calculate long-term debt to total assets.

Here's Lowe's balance sheet with these four financials outlined:

Lowe's Companies, Inc. Consolidated Balance Sheets (In millions, except par value)			
	February 2, 2024		February 3, 2023
Assets			
Current assets:			
Cash and cash equivalents	\$	921	\$ 1,348
Short-term investments		307	384
Merchandise inventory - net		16,894	18,532
Other current assets		949	1,178
Total current assets		19,071	21,442
Property, less accumulated depreciation		17,653	17,567
Operating lease right-of-use assets		3,733	3,518
Long-term investments		252	121
Deferred income taxes - net		248	250
Other assets		835	810
Total assets	\$	41,795	\$ 43,708
Liabilities and shareholders' deficit			
Current liabilities:			
Short-term borrowings	\$	—	\$ 499
Current maturities of long-term debt		517	585
Current operating lease liabilities		487	522
Accounts payable		8,704	10,524
Accrued compensation and employee benefits		954	1,109
Deferred revenue		1,408	1,603
Income taxes payable		33	1,181
Other current liabilities		3,445	3,488
Total current liabilities		15,568	19,511
Long-term debt, excluding current maturities		35,384	32,876
Noncurrent operating lease liabilities		3,737	3,512
Deferred revenue - Lowe's protection plans		1,225	1,201
Other liabilities		931	862
Total liabilities		36,845	57,962
Commitments and contingencies			
Shareholders' deficit:			
Preferred stock - \$5 par value: Authorized - 5.0 million shares; Issued and outstanding - none		—	—
Common stock - \$0.50 par value: Authorized - 5.6 billion shares; Issued and outstanding - 574 million and 601 million, respectively		287	301
Accumulated deficit		(15,637)	(14,862)
Accumulated other comprehensive income		300	307
Total shareholders' deficit		(15,050)	(14,254)
Total liabilities and shareholders' deficit	\$	41,795	\$ 43,708

See accompanying notes to consolidated financial statements.

Source: [Lowe's \(LOW\) 10-K Annual Statement](#) (Balance Sheet)

Thus, in FY 2024, Lowe's reported current assets of \$19,071M, total assets of \$41,795M, current liabilities of \$15,568M, and long-term debt of \$35,384M.

Cash Flow Statement

The cash flow statement outlines the inflows and outflows of cash and cash equivalents, categorizing them into operating, investing, and financing activities over a specific period.

- **Related:** [How to Effectively Read and Analyze a Cash Flow Statement](#)

From the cash flow statement, you'll need to locate these two financials:

- **Cash From Operations (CFO):** To evaluate whether it's positive and for the cash return on assets calculation.
- **Common Stock Issuance:** To evaluate whether the company issued common stock for cash.

Here's Lowe's cash flow statement with these two financials outlined:

Cash flows from operating activities:				
Net earnings	\$	7,726	\$	6,437
Adjustments to reconcile net earnings to net cash provided by operating activities:				
Depreciation and amortization		1,923		1,981
Noncash lease expense		499		530
Deferred income taxes		6		(239)
Asset impairment and loss on property - net		83		2,118
(Gain) loss on sale of business		(79)		421
Share-based payment expense		210		223
Changes in operating assets and liabilities:				
Merchandise inventory - net		1,637		(2,594)
Other operating assets		182		56
Accounts payable		(1,820)		(549)
Deferred revenue		(170)		(183)
Other operating liabilities		(2,057)		388
Net cash provided by operating activities		8,140		8,589
Cash flows from investing activities:				
Purchases of investments		(1,785)		(1,189)
Proceeds from sale/maturity of investments		1,722		1,174
Capital expenditures		(1,964)		(1,829)
Proceeds from sale of property and other long-term assets		53		45
Proceeds from sale of business		100		491
Other - net		(27)		(1)
Net cash used in investing activities		(1,901)		(1,309)
Cash flows from financing activities:				
Net change in commercial paper		(499)		499
Net proceeds from issuance of debt		2,983		9,667
Repayment of debt		(601)		(867)
Proceeds from issuance of common stock under share-based payment plans		141		121
Cash dividend payments		(2,531)		(2,370)
Repurchases of common stock		(6,138)		(14,124)
Other - net		(21)		(5)
Net cash used in financing activities		(6,666)		(7,049)
Effect of exchange rate changes on cash		—		(16)
Net (decrease)/increase in cash and cash equivalents		(427)		215
Cash and cash equivalents, beginning of year		1,348		1,133
Cash and cash equivalents, end of year		\$ 921		\$ 1,348

See accompanying notes to consolidated financial statements.

Source: [Lowe's \(LOW\) 10-K Annual Statement \(Cash Flow Statement\)](#)

Thus, in FY 2024, Lowe's reported cash from operations (CFO) of \$8,140M and proceeds from common stock issuance of \$141M.

To determine the EQ_OFFER factor for the Piotroski F-Score, it's best to use the cash flow statement. The cash flow statement provides the most direct and reliable information about whether a company raised capital through the issuance of common stock during the period, as it shows the actual cash proceeds received. Looking at the balance sheet or shares outstanding count (i.e., on the income statement) could be misleading due to non-cash events or transactions.

In this case, Lowe's states the common stock proceeds come from share-based payment plans, which are typically used to incentivize and reward employees, rather than to raise capital from the market due to financial distress or capital needs. For the purpose of calculating the Piotroski F-Score, specifically the EQ_OFFER factor, the proceeds from common stock issuance should be considered \$0M, not \$141M, since the issuance was not done to raise capital. This would result in an EQ_OFFER score of 1 instead of 0.

Step #2: Calculate Relevant Ratios

Now that we've gathered the relevant financials for the Piotroski F-Score, the next step is to calculate the ratios necessary for completing the Piotroski F-Score calculation.

Profitability

For the profitability category, we'll need to calculate return on assets (ROA) and cash return on assets.

The ROA formula is shown below:

Return on Assets (ROA) = Net Income / Total Assets

To determine the binary score for the change in the ROA (F_ΔROA), we need to compare the current year's ROA to the previous year's. To make this comparison, we'll calculate the ROA for Lowe's for both FY 2023 and FY 2024, as demonstrated below:

ROA FY 2023 [LOW] = \$6,437M / \$43,708M --> **0.147 or 14.7%**

ROA FY 2024 [LOW] = \$7,726M / \$41,795M --> **0.185 or 18.5%**

ROA increased from **14.7%** to **18.5%** from FY 2023 to FY 2024, indicating improved efficiency and profitability.

Next, we need to calculate the cash return on assets, as shown in the formula below:

Cash Return on Assets = Cash Flow From Operations (CFO) / Total Assets

Here's the FY 2024 cash return on assets calculation for Lowe's:

Cash Return on Assets FY 2024 [LOW] = \$8,140M / \$41,795M --> **0.195 or 19.5%**

A cash return on assets of **19.5%** in FY 2024 shows strong cash-generating ability, with the company generating 19.5 cents of cash from operations for every dollar of assets.

Leverage, Liquidity, and Source of Funds

Next, for the leverage, liquidity, and source of funds category, we'll need to calculate long-term debt to total assets (aka the leverage ratio) and the current ratio.

The long-term debt to total assets formula is shown below:

Long-Term Debt to Total Assets = Long-Term Debt / Total Assets

To determine the binary score for the change in the leverage ratio (F_ΔLEVER), we need to compare the current year's leverage ratio to the previous year's. To make this comparison, we'll calculate the leverage ratio for Lowe's for both FY 2023 and FY 2024, as demonstrated below:

Long-Term Debt to Total Assets FY 2023 [LOW] = \$32,876M / \$43,708M --> **0.752 or 75.2%**

The long-term debt to total assets ratio increased from **75.2%** to **84.7%** between FY 2023 and FY 2024, suggesting higher financial risk and a potential increase in interest expenses.

Next, we need to calculate the current ratio, as shown in the formula below:

Current Ratio = Current Assets / Current Liabilities

To determine the binary score for the change in the current ratio ($F_ALIQUID$), we need to compare the current year's current ratio to the previous year's. To make this comparison, we'll calculate the current ratio for Lowe's for both FY 2023 and FY 2024, as demonstrated below:

Current Ratio FY 2023 [LOW] = \$21,442M / \$19,511M --> **1.10**

Current Ratio FY 2024 [LOW] = \$19,071M / \$15,568M --> **1.23**

The current ratio improved from **1.10** to **1.23** from FY 2023 to FY 2024, indicating stronger short-term liquidity, with more current assets available to cover current liabilities.

Operating Efficiency

Finally, for the operating efficiency category, we'll need to calculate the gross profit margin (GPM) and asset turnover ratio.

The GPM ratio is shown below:

Gross Profit Margin (GPM) = Gross Profit / Total Revenues

To determine the binary score for the change in the GPM (F_MARGIN), we need to compare the current year's GPM to the previous year's. To make this comparison, we'll calculate the GPM for Lowe's for both FY 2023 and FY 2024, as demonstrated below:

Gross Profit Margin FY 2023 [LOW] = \$32,257M / \$97,059M --> **0.332 or 33.2%**

Gross Profit Margin FY 2024 [LOW] = \$28,844M / \$86,377M --> **0.334 or 33.4%**

The GPM increased slightly from **33.2%** to **33.4%** between FY 2023 and FY 2024, suggesting maintained or slightly improved pricing power and cost control.

Next, we need to calculate the asset turnover ratio, as shown in the formula below:

Asset Turnover Ratio = Total Revenues / Total Assets

To determine the binary score for the change in the asset turnover ratio (F_TURN), we need to compare the current year's asset turnover ratio to the previous year's. To make this comparison, we'll calculate the asset turnover ratio for Lowe's for both FY 2023 and FY 2024, as demonstrated below:

Asset Turnover Ratio FY 2023 [LOW] = \$97,059M / \$43,708M --> **2.22**

Asset Turnover Ratio FY 2024 [LOW] = \$86,377M / \$41,795M --> **2.07**

The asset turnover ratio decreased from **2.22** to **2.07** from FY 2023 to FY 2024, indicating a slight decline in operational efficiency, with less revenue generated per dollar of assets.

Step #3: Determine Piotroski's F-Score

Now that we've gathered the relevant financials and calculated the ratios needed for the Piotroski F-Score, the final step is to determine the final score for Lowe's, in this case for FY 2024.

Profitability

- 1. **Positive Return on Assets (F_ROA):** Lowe's reported a ROA of 18.5% in FY 2024, which is positive. $F_ROA = 1$.
- 2. **Positive Operating Cash Flow (F_CFO):** Lowe's operating cash flow in FY 2024 was \$8,140M, which is also positive. $F_CFO = 1$.
- 3. **Improving Return on Assets (F_ARO):** Lowe's ROA increased from 14.7% in FY 2023 to 18.5% in FY 2024, which is an improvement. $F_MARGIN = 1$.
- 4. **Quality of Earnings ($F_ACCRUAL$):** Lowe's cash return on assets of 19.5% is greater than its ROA of 18.5% in FY 2024. $F_ACCRUAL = 1$.

Leverage, Liquidity, and Source of Funds

- 5. **Decreasing Leverage (F_ALEVER):** Lowe's long-term debt to total assets ratio increased from 75.2% in FY 2023 to 84.7% in FY 2024, which means the company is more levered. $F_ALEVER = 0$.
- 6. **Increasing Liquidity ($F_ALIQUID$):** Lowe's current ratio increased from 1.10 in FY 2023 to 1.23 in FY 2024, which is an improvement. $F_ALIQUID = 1$.
- 7. **No Equity Issuance (EQ_OFFER):** Lowe's did not issue new common stock to raise capital in FY 2024. $EQ_OFFER = 1$.

Operating Efficiency

- 8. **Improving Gross Margin (F_MARGIN):** Lowe's gross margins increased from 33.2% in FY 2023 to 33.4% in FY 2024, which is a slight improvement. $F_MARGIN = 1$.
- 9. **Improving Asset Turnover (F_TURN):** Lowe's asset turnover ratio decreased from 2.22 to 2.07, which suggests the company is not using its assets efficiently. $F_TURN = 0$.

By summing the individual scores, we can calculate Lowe's Piotroski F-Score in FY 2024 as follows:

F_SCORE FY 2024 [LOW] = 1 (F_ROA) + 1 (F_CFO) + 1 (F_ARO) + 1 ($F_ACCRUAL$) + 0 (F_ALEVER) + 1 ($F_ALIQUID$) + 1 (EQ_OFFER) + 1 (F_MARGIN) + 0 (F_TURN) --> **7**

characteristics, albeit with some limitations in leverage and asset turnover.

Piotroski's F-Score and Investment Performance

To gain a comprehensive understanding of the Piotroski F-Score, let's first explore Piotroski's original findings and then delve into the insights provided by subsequent independent research on the effectiveness and robustness of this investment strategy.

Piotroski's Research

Piotroski developed the F-Score as a method to identify financially strong companies within the high book-to-market (value) segment of the stock market.

Piotroski backtested the F-Score over a 20-year period from 1976 to 1996 in his original paper and found impressive results, as detailed below.

Outperforming the Market

Piotroski discovered that by investing in value stocks, characterized by low market prices relative to their fundamental value as indicated by high book-to-market ratios, and selecting those with the highest F-Scores (8 or 9), generated an average annual return of 13.4%.

In comparison, the average return for all value stocks was only 5.9% per year during the same period.

This means high F-Score value stocks outperformed the average value stock by 7.5% annually (13.4% - 5.9%).

Long-Short Strategy Performance

The performance of a long-short strategy that involved buying high F-Score companies (8 or 9) and short-selling low F-Score companies (0 or 1) was even more noteworthy. This approach generated an average annual return that outperformed the market by 23.0% over the same period.

We've outlined these first two findings in the table below (taken from Piotroski's paper):

Panel A: One-Year Market-Adjusted Returns ^a								
	Mean	10%	25%	Median	75%	90%	%Positive	n
All Firms	0.059	-0.560	-0.317	-0.061	0.255	0.708	0.437	14,043
F_SCORE								
0	-0.061	-0.710	-0.450	-0.105	0.372	0.766	0.386	57
1	-0.102	-0.796	-0.463	-0.203	0.087	0.490	0.307	339
2	-0.020	-0.686	-0.440	-0.151	0.198	0.732	0.374	859
3	-0.015	-0.691	-0.411	-0.142	0.186	0.667	0.375	1618
4	0.026	-0.581	-0.351	-0.100	0.229	0.691	0.405	2462
5	0.053	-0.543	-0.307	-0.059	0.255	0.705	0.438	2787
6	0.112	-0.493	-0.278	-0.024	0.285	0.711	0.471	2579
7	0.116	-0.466	-0.251	-0.011	0.301	0.747	0.489	1894
8	0.127	-0.462	-0.226	0.003	0.309	0.710	0.504	1115
9	0.159	-0.459	-0.265	-0.012	0.327	0.885	0.486	333
Low Score	-0.096	-0.781	-0.460	-0.200	0.107	0.548	0.318	396
High Score	0.134	-0.462	-0.236	0.000	0.316	0.757	0.500	1448
High—All	0.075	0.098	0.081	0.061	0.061	0.049	0.063	—
t-stat/(p-value)	3.140	—	—	(0.000)	—	—	(0.000)	—
Bootstrap Rslt (p-value)	2/1000 (0.002)	0/1000 (0.000)	0/1000 (0.000)	0/1000 (0.000)	2/1000 (0.002)	126/1000 (0.126)	—	—
High—Low	0.230	0.319	0.224	0.200	0.209	0.209	0.182	—
t-stat/(p-value)	5.590	—	—	(0.000)	—	—	(0.000)	—
Bootstrap Rslt (p-value)	0/1000 (0.000)	0/1000 (0.000)	0/1000 (0.000)	0/1000 (0.000)	0/1000 (0.000)	18/1000 (0.018)	—	—

Source: [Value Investing: The Use of Historical Financial Statement Information to Separate Winners from Losers](#) (pg. 16)

Higher Proportion of Winning Stocks

Piotroski's research also revealed that the high F-Score portfolio, consisting of value stocks with scores of 8 or 9, had a significantly higher proportion of winning stocks compared to the average value portfolio:

- **High F-Score Portfolio Performance:** The high F-Score portfolio (scores 8 or 9) successfully picked winning stocks 50% of the time.
- **Overall Value Stock Portfolio Performance:** The portfolio comprising all value stocks showed a winner percentage of 43.7%.
- **Low F-Score Portfolio Performance:** The low F-Score portfolio (scored 0 or 1) had the least impressive performance, with only 31.8% of stocks being winners.

These differences in proportions were statistically significant based on a binomial test. Therefore, this finding suggests that the F-Score effectively identifies value stocks with a higher likelihood of generating positive returns, thereby increasing the overall success rate of the portfolio.

Performance Across Company Sizes

Piotroski also examined the effectiveness of the F-Score strategy across different company sizes. He divided the value stock universe into three categories: **small**, **medium**, and **large-sized** firms.

- **Small-Sized Firms:** The high F-Score portfolio generated an average annual return of **17.9%**, outperforming the low F-Score portfolio by **27.0%** per year.
- **Medium-Sized Firms:** The high F-Score portfolio achieved an average annual return of **7.9%**, surpassing the low F-Score portfolio by **17.3%** annually.
- **Large-Sized Firms:** The high F-Score portfolio of large companies outperformed the low F-Score portfolio by only **1.7%** per year, and the difference in returns was **not statistically significant**.

We've outlined these findings in the table below (also taken from Piotroski's paper):

Table 4: One-Year Market-Adjusted Buy-and-Hold Returns to a Value Investment Strategy Based on Fundamental Signals by Size Partition

	Small Firms			Medium Firms			Large Firms		
	Mean	Median	n	Mean	Median	n	Mean	Median	n
All Firms	0.091	-0.077	8302	0.008	-0.059	3906	0.003	-0.028	1835
F SCORE									
0	0.000	-0.076	32	-0.146	-0.235	17	-0.120	-0.047	8
1	-0.104	-0.227	234	-0.083	-0.228	79	-0.136	-0.073	26
2	-0.016	-0.171	582	-0.045	-0.131	218	0.031	-0.076	59
3	0.003	-0.168	1028	-0.049	-0.108	429	-0.036	-0.068	161
4	0.058	-0.116	1419	-0.024	-0.104	687	-0.002	-0.023	356
5	0.079	-0.075	1590	0.028	-0.060	808	-0.004	-0.031	389
6	0.183	-0.030	1438	0.029	-0.041	736	0.012	-0.004	405
7	0.182	0.005	1084	0.027	-0.028	540	0.028	-0.015	270
8	0.170	0.001	671	0.081	0.024	312	0.012	-0.041	132
9	0.204	-0.017	224	0.068	0.032	80	0.059	-0.045	29
Low Score	-0.091	-0.209	266	-0.094	-0.232	96	-0.132	-0.066	34
High Score	0.179	-0.007	895	0.079	0.024	392	0.020	-0.045	161
High-All	0.088	0.070	—	0.071	0.083	—	0.017	-0.017	—
t-statistic/(p-value)	2.456	(0.000)	—	2.870	(0.000)	—	0.872	(0.203)	—
High-Low	0.270	0.202	—	0.173	0.256	—	0.152	0.021	—
t-statistic/(p-value)	4.709	(0.000)	—	2.870	(0.000)	—	1.884	(0.224)	—

Note: Each year, all firms on COMPUSTAT with sufficient size and BM data are ranked on the basis of the most recent fiscal year-end market capitalization. The 33.3 and 66.7 percentile cutoffs from the prior year's distribution of firm size (MVE) are used to classify the high BM firms into small, medium, and large firms each year. All other definitions and test statistics are as described in table 3.

Source: [Value Investing: The Use of Historical Financial Statement Information to Separate Winners from Losers](#) (pg. 19)

In summary, this evidence implies that the F-Score strategy is most effective when applied to small and medium-sized value companies, where market inefficiencies and mispricing may be more prevalent. The lower efficacy among large companies could be attributed to greater market efficiency and more extensive analyst coverage, which may result in stock prices more accurately reflecting the company's financial strength.

Other Independent Research

Since Piotroski's original publication, numerous researchers have conducted independent studies to validate the effectiveness of the F-Score and explore its applicability in various contexts. These studies provide valuable insights into the robustness of the F-Score and its potential for identifying financially strong companies across different markets, sectors, and investment strategies.

Equities Lab: Piotroski F-Score Backtest

[Equities Lab](#) evaluated the Piotroski F-Score in an article and performed a backtest from **2004 to 2024** (span of **20 years**), screening for scores greater than 7. With 315 matches, here's what they discovered:

- Companies with a perfect F-Score (9) outperformed the S&P 500 over a 20-year period, generating an annual return of **13%** compared to the S&P 500's **9.92%**. However, these perfect-score companies also exhibited higher volatility.
- Companies with an F-Score of 7 or greater slightly outperformed the S&P 500 over the same period, with an annual return of **9.95%** compared to the S&P 500's **9.92%**.
- Companies with an F-Score of 5 or less significantly underperformed the S&P 500, generating a total return of **190.2%** compared to the S&P 500's **574.3%**.
- The Piotroski F-Score proves to be an effective predictor of a company's future performance, with lower scores generally indicating worse performance and higher scores indicating better performance.
- The F-Score works well alongside other factors such as momentum and value (e.g., price to earnings (P/E) ratio), except when the P/E is over 60x.
- The effectiveness of the F-Score varies across different sectors, with some sectors showing a clearer trend between F-Score and performance than others.

The article concludes that the Piotroski F-Score is a reliable indicator of a company's overall health and future direction. It demonstrates the effectiveness of the F-Score in predicting performance across various market scenarios and in conjunction with other factors. However, the article also highlights that the F-Score's effectiveness may vary depending on the sector and that investors should consider portfolio diversification to mitigate the risk of individual stock "faceplants."

Alpha Architect: Creating a Better F-Score

The article "[Creating a Better F-Score](#)" by Alpha Architect discusses potential improvements to the Piotroski F-Score. The authors propose a modified version called the **Financial Strength Score (FS-Score)**, which incorporates slight changes to the original F-Score while maintaining its simplicity.

current profitability, stability, and recent operational improvements.

- **Current Profitability:** The authors replace the cash flow from operations (CFO) variable with free cash flow divided by total assets (FCFTA), to account for the impact of capital expenditures on a stock's cash flows.
- **Stability:** Replaces the F-Score's equity issuance variable (EQ_OFFER) with net equity issuance (NEQISS), defined as repurchases minus issuances, to better capture the relative size of both buybacks and issuances.
- **Recent Operational Improvements:** Introduced to assess whether the business has operational momentum, focusing on improvements in ROA, FCFTA, gross margin ratio, and asset turnover ratio.
- Backtesting results from **1974 to 2014** (span of **50 years**), show that the FS-Score outperforms the F-Score by a small but economically meaningful amount, with a compound annual growth rate (CAGR) of **13.3%** compared to **12.6%** for the F-Score. Both also beat the CAGR of **11.2%** for the S&P 500 during the same period.
- When applied to cheap stocks (defined as the 20% cheapest based on book-to-market), the FS-Score features a CAGR of **15.9%** compared to **15.2%** for the F-Score.

The article concludes that the small tweaks applied to the F-Score result in a more intuitive and effective financial strength measure, the FS-Score, which outperforms the original F-Score in both the overall market and the value stock universe. The authors suggest that the FS-Score is more grounded in value-investing philosophy than the F-Score, making it a potentially valuable tool for investors seeking to identify financially robust stocks.

Utility of Piotroski F-Score for Predicting Growth-Stock Returns

"[Utility of Piotroski F-Score for Predicting Growth-Stock Returns](#)," by Jan-Hendrik Markus Mohr examines the utility of the Piotroski F-Score in predicting returns for growth stocks in the Eurozone equity market from **1999 to 2010** (span of **12 years** according to the paper). The study aims to contribute empirical evidence to the ongoing debate on whether fundamental analysis is dependent on the valuation context of the sampled stocks.

Key findings of the paper include:

- A market-neutral strategy of buying high F-Score growth stocks and shorting low F-Score growth stocks generates a significant annual return of **24.57%** over the twelve-year period from 1999 to 2010.
- High F-Score growth stocks outperform the market consistently, with an average annual market-adjusted return of **10.74%**, while low F-Score growth stocks underperform the market with an average annual market-adjusted return of **-13.82%**.
- The median market capitalization and trading volume of high F-Score growth stocks are generally higher than those of the entire growth stock universe and low F-Score growth stocks, suggesting potential implementation challenges for shorting low F-Score stocks.
- When controlling for other sources of cross-sectional variation in returns (size, price-to-book (P/B), momentum, accruals, and equity offerings), the F-Score remains a significant explanatory factor for market-adjusted returns, increasing the R^2 of the regression model.
- The findings support the applicability of the Piotroski F-Score in predicting returns for growth stocks in the Eurozone market, challenging the view that fundamental analysis is strongly context-dependent and loses its predictive ability when applied outside the original value stock universe.

The paper concludes by discussing potential ways to implement the F-Score-based strategy in a real-life setting, such as through a hedge fund, mutual fund, or a long-short equity strategy, while acknowledging the limitations and challenges associated with shorting low F-Score growth stocks.

Predicting Firms' Financial Distress

Lastly, "[Predicting Firms' Financial Distress: An Empirical Analysis Using the F-Score Model](#)," by Rahman, Sa, and Masud (2021) explores the applicability of the Piotroski F-Score in predicting financial distress for publicly listed companies in the United States. The study aims to determine whether the F-Score and its individual components can effectively identify firms at high risk of default.

Key findings of the paper include:

- The Piotroski F-Score has a significant relationship with the probability of financial distress. Companies with lower F-Scores are more likely to experience financial distress compared to those with higher F-Scores.
- Distressed firms typically record negative cash flow from operations (CFO) in the year before default, while non-distressed firms generally maintain positive CFO.
- Firms in financial distress experience a significantly larger decline in return on assets (ROA) compared to non-distressed firms in the year prior to distress.
- The study compares two models: Model 1, which uses the aggregate F-Score, and Model 2, which employs the individual components of the F-Score. The results show that Model 1 has a higher predictive accuracy (**79.2%**) compared to Model 2 (**72.9%**) when applied to the hold-out sample.
- The research provides evidence supporting the use of the Piotroski F-Score and its components in predicting financial distress for U.S. public companies, contributing to the existing literature on the F-Score's applicability.

In summary, the key takeaways are that a low Piotroski F-Score, negative cash flow from operations, and a sharp decline in ROA are significant indicators that a firm is likely to experience financial distress in the near future. The aggregate F-Score is an effective tool for predicting this distress.

Limitations of the Piotroski F-Score

While the Piotroski F-Score is a popular financial scoring model for evaluating a company's financial health, it's important to understand its limitations before considering its application.

To begin, in his original paper, Piotroski acknowledged the scope and intent of the F-Score:

"Although this paper does not purport to find the optimal set of financial ratios for evaluating the performance prospects of individual "value" firms, the results convincingly demonstrate that investors can use relevant historical information to eliminate firms with poor future prospects from a generic high BM portfolio."

- Joseph D. Piotroski in "Value Investing: The Use Of Historical Financial Statement Information To Separate Winners From Losers," pg. 33

Piotroski's statement highlights the necessity of understanding the F-Score's boundaries and incorporating it as a component within a comprehensive investment analysis strategy, rather than relying on it in isolation.

Now, below are the key limitations of the Piotroski F-Score that investors should keep in mind before using the model:

Reliance on Historical Financial Data

One significant limitation of the Piotroski F-Score, as noted by Piotroski himself, is its dependency on historical financial data, which may not always predict future company performance accurately. As noted by Piotroski:

"One limitation of this study is the existence of a potential data-snooping bias. The financial signals used in this paper are dependent, to some degree, on previously documented results; such a bias could adversely affect the out-

- Joseph D. Piotroski in "Value Investing: The Use Of Historical Financial Statement Information To Separate Winners From Losers," pg. 35

Here, Piotroski highlights the challenge of relying on historical financial data to predict future outcomes, recognizing the importance of considering the limitations inherent in using past financial performance to forecast future returns.

Sensitivity to Small Changes in Financial Metrics

The Piotroski F-Score is sensitive to small changes in the underlying financial metrics. As the F-Score is based on binary criteria, even a slight change in a company's financial ratios can impact the overall score.

For example, if a company's return on assets (ROA) increases from 0.9% to 1.1%, the F-Score would award a point for the positive change, despite the relatively small improvement. Conversely, if the ROA decreases from 1.1% to 0.9%, the company would lose a point, even though the decline is minimal.

This sensitivity to small changes can lead to a company's F-Score fluctuating from year to year, even if its overall financial health remains relatively stable. As a result, investors may need to exercise caution when interpreting changes in a company's F-Score and consider the magnitude of the changes in the underlying financial metrics.

To mitigate this limitation, investors could consider using a more gradual scoring system or establishing materiality thresholds for changes in financial ratios. By doing so, the F-Score would be less sensitive to minor fluctuations and provide a more robust assessment of a company's financial health over time.

Short-Term Orientation

The F-Score's focus on annual changes in financial metrics may overlook longer-term trends or strategic shifts that could impact a company's performance, as detailed below:

- **Long-Term Investments:** Companies making significant investments in research and development (R&D), infrastructure, or acquisitions may experience short-term declines in financial metrics, which could negatively impact their F-Score. However, these investments may lead to improved long-term performance and competitive advantages.
- **Business Cycle Sensitivity:** Some companies may exhibit cyclical financial performance due to the nature of their industry or the economy. The F-Score's short-term focus may not capture these cyclical trends, leading to an inaccurate assessment of a company's long-term financial health.

To address this limitation, investors should consider supplementing the F-Score with an analysis of longer-term financial trends, industry-specific cycles, and the company's strategic initiatives to gain a more comprehensive view of its potential future performance.

Lack of Qualitative Factors

The F-Score does not directly incorporate qualitative aspects such as management quality, competitive advantages, or market trends, which can significantly influence a company's success.

- **Management Effectiveness:** A company's financial performance is often driven by the quality of its [management team](#). Factors such as the management's track record, experience, and strategic vision can have a substantial impact on a company's long-term success, but are not captured by the F-Score.
- **Competitive Landscape:** The F-Score does not account for the strength of a company's competitive position within its industry. Factors such as market share, brand recognition, and barriers to entry can significantly influence a company's ability to generate sustainable profits and growth.
- **Macroeconomic and Industry Trends:** Changes in the broader economic environment or industry-specific trends can have a significant impact on a company's financial performance. The F-Score does not directly incorporate these external factors, which may limit its predictive power during periods of significant change or disruption.

To mitigate this limitation, investors should conduct a thorough qualitative analysis alongside the F-Score, considering factors such as the strength of the management team, the company's competitive position, and relevant market and industry trends. By combining quantitative and qualitative insights, investors can make more informed decisions about a company's overall prospects.

Market Efficiency

The degree of market efficiency may influence the effectiveness of the F-Score. Piotroski acknowledged that further research is needed to determine whether the market behavior documented in his paper is a result of inefficiency or a rational pricing strategy:

"Whether the market behavior documented in this paper equates to inefficiency, or is the result of a rational pricing strategy that only appears to be anomalous, is a subject for future research."

- Joseph D. Piotroski in "Value Investing: The Use Of Historical Financial Statement Information To Separate Winners From Losers," pg. 34

This uncertainty surrounding market efficiency suggests that investors should be aware of the potential limitations of the F-Score and monitor its effectiveness over time, as the strategy's success may diminish if the market becomes more efficient in pricing value stocks.

Accounting Manipulations

The F-Score's reliance on reported financial data makes it vulnerable to potential accounting manipulations or irregularities, which could distort the score's accuracy.

In other words, if companies engage in aggressive revenue recognition, expense understatement, or other misleading accounting practices, it could artificially inflate their F-Score and provide an inaccurate picture of their financial health.

Investors should therefore consider additional due diligence, such as analyzing footnotes and other qualitative information, to identify potential red flags in a company's financial reporting.

Industry-Specific Factors

Lastly, the F-Score may not be equally effective across all sectors due to variations in financial characteristics and reporting standards. Some examples include:

- **Capital-Intensive Industries:** Sectors like manufacturing or utilities may have different leverage and asset turnover ratios that require contextual evaluation.
- **Service-Based Businesses:** These companies may prioritize metrics such as revenue growth and profit margins over asset turnover.
- **Highly-Regulated Industries:** Financial services and healthcare sectors often have unique reporting standards that demand a more nuanced approach to the F-Score.

Why Piotroski's F-Score No Longer Works

Building upon the limitations discussed above, a recent study by titled "[Why Piotroski's F-Score No Longer Works](#)" provides empirical evidence suggesting that the Piotroski F-Score has failed to deliver similar returns in the years following its publication. The study analyzed the performance of the F-Score strategy from **1998 to 2021** (span of **23 years**) and found the following:

Key findings of the article include:

- **Poor Out-of-Sample Performance:** When applying the same criteria used in Piotroski's original study, a strategy of buying high F-Score firms and shorting low F-Score firms would have generated an average annual loss of **-9.53%** over the last ten years and **-11.75%** over the last 20 years.
- **Flawed Stock Selection Criteria:** The article argues that Piotroski's focus on high book-to-market (BM) firms is problematic, as this ratio has little bearing on whether a company is mispriced. Additionally, the exclusion of financial firms due to the lack of current ratio data may have skewed the results.
- **Ineffective Individual Factors:** When examining the individual components of the F-Score, the author found that only one factor, accruals, showed any predictive power in separating good high book-to-market firms from bad ones since 1999.
- **Questionable Research Methodology:** The author points out that Piotroski's research method, which involved buying stocks four months after the end of the fiscal year based on the book-to-market ratio at that time, would not be practical for actual investors. Additionally, the omission of the years 1997 to 1999 from the original study raises concerns about the robustness of the results.
- **Flawed Financial Metrics:** The article argues that the book-to-market ratio and the debt-to-assets ratio, two central components of the F-Score, are inadequate measures for assessing a company's intrinsic value and financial health.

The author proposes an alternative scoring system, the "**Y-Score**," which modifies Piotroski's original factors and incorporates additional value ratios (see the article for specifics). While the Y-Score appears to perform better than the F-Score in the period from **1999 to 2020** (span of **21 years**), it still fails to generate consistent outperformance in the last ten years.

The article concludes by highlighting the lessons learned from Piotroski's out-of-sample failure, including the limitations of backtesting, the importance of disclosing research methodology, and the need for researchers to have practical investing experience. The article also emphasizes the shortcomings of relying on one-year changes in financial ratios and the importance of considering industry-specific factors when analyzing a company's financial health.

The Bottom Line

The Piotroski F-Score is a financial scoring model developed by Joseph Piotroski in 2000 to identify financially strong companies within the high book-to-market (value) segment of the stock market. The F-Score is based on nine criteria related to a company's profitability; leverage, liquidity, source of funds; and operating efficiency. Each criterion is assigned a binary score of 1 or 0, and the sum of these scores determines the final F-Score, which ranges from 0 to 9. A higher F-Score indicates better financial health.

Piotroski's original study (1976 to 1996) found that high F-Score value stocks outperformed the average value stock by 7.5% annually, and a long-short strategy based on F-Score generated a 23.0% annual return above the market. The F-Score worked best for small and medium-sized value companies.

Subsequent studies by Alpha Architect, Equities Lab, Jan-Hendrik Markus Mohr, and Rahman et al. confirmed the F-Score's effectiveness in identifying financially strong companies that outperform the market, across various market segments and regions.

However, the F-Score has limitations, such as potential ineffectiveness across sectors, vulnerability to accounting manipulations, lack of qualitative factor consideration, and reliance on historical and short-term data. A recent study, "Why Piotroski's F-Score No Longer Works," highlights the F-Score's diminished effectiveness in recent years due to poor out-of-sample performance, flawed stock selection criteria, ineffective individual factors, questionable research methodology, and flawed financial metrics.

Thus, the F-Score's individual components and overall scoring system have failed to deliver consistent results when applied across different time periods. This serves as a reminder for investors to be cautious when interpreting academic papers, as many researchers lack practical investing experience. Piotroski's main contribution was demonstrating that fundamental accounting data could help identify promising opportunities among value stocks, which are often overlooked by investors who focus primarily on growth or momentum strategies. While the F-Score provided academic validation for strategies that may have already been employed by some investors, viewing it as a definitive blueprint for stock picking may lead to disappointing outcomes.